

GranTourisma AutoHaus — Full Insights Report

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Comprehensive SQL-Driven Analysis of Dealership Performance

1. Introduction

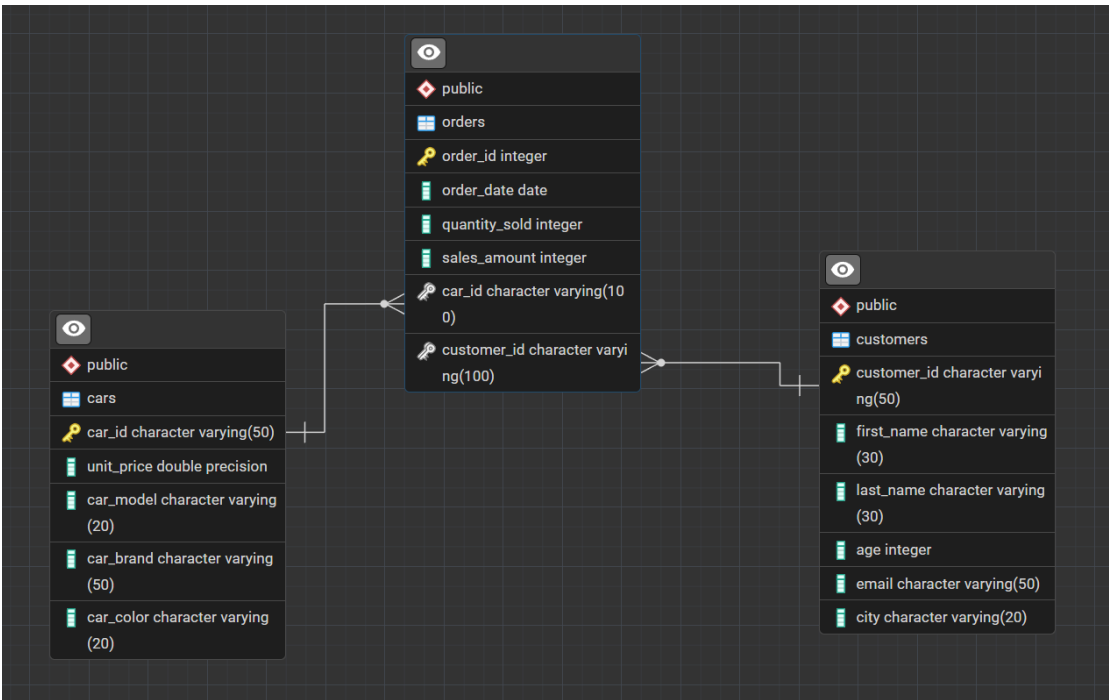
This report presents a full analytical breakdown of GranTourisma AutoHaus, a car dealership used to demonstrate real-world SQL analytics. The analysis is structured into five pillars, each addressing a core business question:

1. Overall business performance
2. Product performance
3. Customer demographics
4. Geographic sales analysis
5. Sales trend analysis

All insights are derived from three relational tables:

- customers
- cars
- orders

2. Database Schema



3. Pillar 1 — Overall Business Performance

Results

- Total Revenue: \$125,714,000
- Total Orders: 1050
- Total Vehicles Sold: 4223
- Average Order Value: \$119,727.62

Insights

1. Strong revenue base driven by high-value transactions

The dealership averages around 119.7k per order, indicating a product mix dominated by mid-to-high-end vehicles. This positions the dealership as a premium retailer rather than a volume-based seller.

2. Healthy sales volume helps consistent revenue generation

Selling 4223 vehicles across 1050 orders means customers frequently purchase multiple units per order, which is very healthy for the business.

3. Revenue concentration risk

The dealership leans heavily on its luxury models, so if demand for high-end cars dips, revenue could take a noticeable hit.

4. Pillar 2 — Product Performance

4.1 Brand Revenue

Top Brands

1. Audi — \$22,122,000
2. Mercedes-Benz — \$18,812,000
3. BMW — \$18,790,000

Insights

1. German luxury brands dominate the dealership's revenue

Audi, Mercedes, and BMW account for approximately \$59M in revenue, contributing nearly half of total sales. This highlights a strong customer preference for luxury European vehicles.

2. Audi outperforms competitors in both value and volume

The Audi A6 generates the highest revenue at \$13.45M, while the Audi A4 records the highest sales volume with 271 units sold. This demonstrates Audi's broad appeal across different customer segments, making it the dealership's most valuable brand.

3. Premium models generate strong revenue despite lower sales volumes

The Genesis G70 recorded only four sales but contributed approximately \$1.6M in revenue. This demonstrates how high-value vehicles can significantly boost revenue even when sales numbers are low.

4. Customer colour preferences follow familiar market trends

White, Black, Blue, and Red are the most popular vehicle colours among customers. In contrast, Gray and Green attract considerably less demand, presenting an opportunity to better align inventory with customer preferences.

5. Pillar 3 — Customer Demographics

5.1 Customer Count by Age Group

- Adults: 63
- Young Adults: 27
- Older Adults: 20

5.2 Revenue by Age Group

- Adults: \$71.3M
- Young Adults: \$30.6M
- Older Adults: \$23.7M

Insights

1. Adults (30–39) are the dealership's core revenue drivers

They represent 57% of customers and generate \$71.3M. Their high engagement suggests strong purchasing power and brand loyalty.

2. Young Adults show strong spending potential

Despite being a smaller group, they generate nearly half the revenue of Adults. Their average spend is nearly identical, indicating strong long-term value.

3. Older Adults remain a stable but smaller segment

Their spending patterns are consistent, but they represent a smaller share of total customers.

4. A small group of high-value customers drives a large share of revenue

The top 10 customers collectively generate over \$20M in revenue, indicating strong repeat purchasing behavior and significant lifetime value.

6. Pillar 4 — Geographic Sales Analysis

6.1 Revenue by City

- **Top:** Tulsa (\$2.64M), Glendale (\$2.38M), Aurora (\$2.31M)
- **Bottom:** Hialeah (\$172k), Birmingham (\$88k)

6.2 Order Volume by City

- **Top:** Aurora (20 orders), Wichita (19), New York (18)

6.3 Luxury Car Orders

- **Top:** Baltimore, Tulsa, Cincinnati (4 each)

Insights

1. Tulsa is the dealership's most valuable market

Despite only 15 orders, Tulsa generates the highest revenue, indicating a strong luxury-car market.

2. Aurora leads in order volume but not revenue

Aurora's high order count suggests strong engagement, but lower average order values indicate mid-range preferences.

3. Luxury vehicle demand is geographically widespread

Baltimore and Cincinnati show unexpectedly strong demand for luxury cars, even though they are not leading cities in total revenue.

4. Low-performing cities highlight areas for expansion

Hialeah and Birmingham are currently underperforming in sales activity, indicating opportunities for focused marketing efforts or adjustments in product availability.

7. Pillar 5 — Sales Trend Analysis

7.1 Monthly Revenue

- **Best Month:** October 2023 (\$12.16M)
- **Worst Month:** June 2024 (\$4.83M)

7.2 Units Sold

- **Peak:** October 2023 (415 units)
- **Lowest:** June 2024 (169 units)

7.3 Yearly Revenue

- **2023:** \$71.8M
- **2024:** \$53.9M

Insights

1. 2023 shows strong, stable growth

Revenue climbs steadily through 2023, peaking in October. This suggests effective inventory management and strong seasonal demand.

2. 2024 begins strongly but declines sharply

Revenue collapses by 45% from May to June 2024. This may indicate inventory shortages, economic shifts, or operational issues.

3. Unit sales mirror revenue trends

The decline in 2024 is driven by lower sales volume, not pricing changes.

4. Year-over-year performance reveals a concerning trend

2024 underperforms 2023 by approximately \$18M, signalling a loss of momentum that requires strategic intervention.

8. Strategic Recommendations

1. Strengthen luxury inventory & partnerships

Given the dominance of Audi, BMW, and Mercedes:

- Increase allocation of high-demand models
- Negotiate better dealership incentives
- Expand test-drive and premium experience offerings

2. Rebalance inventory based on colour demand

- Reduce stock of Gray and Green vehicles
- Increase White, Black, Blue, and Red inventory
- Offer custom-order options for low-demand colours

3. Target Adults (30–39) with personalised marketing

- Loyalty programs
- Exclusive upgrade offers
- Early access to new models
- Financing incentives

4. Develop a Young Adult growth strategy

This group shows strong spending potential. Introduce:

- Entry-level luxury financing
- Subscription models
- First-time buyer incentives

5. Investigate 2024 performance decline

Conduct a root-cause analysis into:

- Inventory shortages
- Sales team performance
- Marketing spend
- Economic conditions
- Competitor activity

6. Expand presence in high-value cities

Prioritise Tulsa, Glendale, Aurora, and Cincinnati for:

- Local events
- Premium inventory
- Targeted digital ads

7. Improve performance in underperforming cities

For Hialeah and Birmingham:

- Increase brand awareness
- Offer promotional pricing
- Partner with local businesses
- Introduce lower-priced models

8. Build a VIP customer program

Given the high lifetime value of top customers:

- Dedicated account managers
- Priority servicing
- Exclusive events
- Early access to new models